



CANDIDATE
NAME

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CIVICS
GROUP

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REGISTRATION
NUMBER

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H2 Biology

9744/03

Paper 3 Structured & Free Response Questions

19 September 2022

2 hours

Candidates answer **Section A** on the Question Paper and **Section B** on the Answer Booklet.

Additional Materials: 12-page Answer Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and registration number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid/tape.

Section A

Answer **all** questions.

Section B

Answer **one** question on the **12-page Answer Booklet**.

Write your answer to each part of the question on a fresh sheet of paper.

The use of an approved scientific calculator is expected, where appropriate.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, ensure that you submit both the question paper and answer booklets.

For Examiner's Use	
Section A	
1	
2	
3	
Section B	
4 OR 5	
Total	75

This document consists of **18** printed pages and **2** blank pages.

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Section A

- 1 (a) Influenza causes occasional pandemics that have claimed the lives of millions of people.

Different strains of influenza are named in terms of 'H' and 'N'. For example, avian influenza is named H5N1.

Fig. 1.1 shows the main structural features of the H5N1 influenza virus.

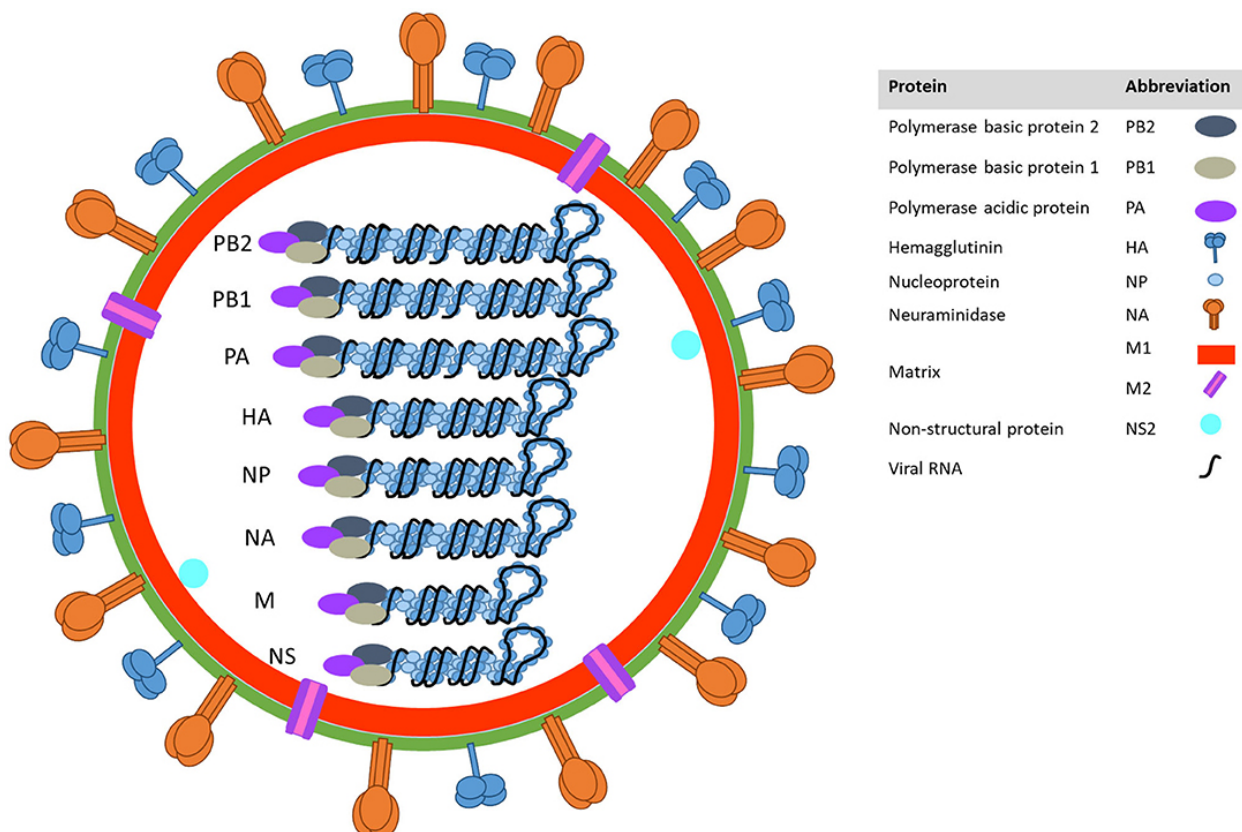


Fig. 1.1

- (i) Explain the functions of haemagglutinin and neuraminidase in an enveloped virus, such as H5N1.

Haemagglutinin

.....
 [1]

Neuraminidase

.....
 [1]

- (ii) Other than components stated in Fig. 1.1, name two other components of the envelope of a virus, such as H5N1.

..... [1]

- (iii) The emergence of new strains of influenza continues to pose challenges to public health and the scientific communities.

Fig. 1.2 shows various ways that new strains of influenza, such as H5N1, may arise.

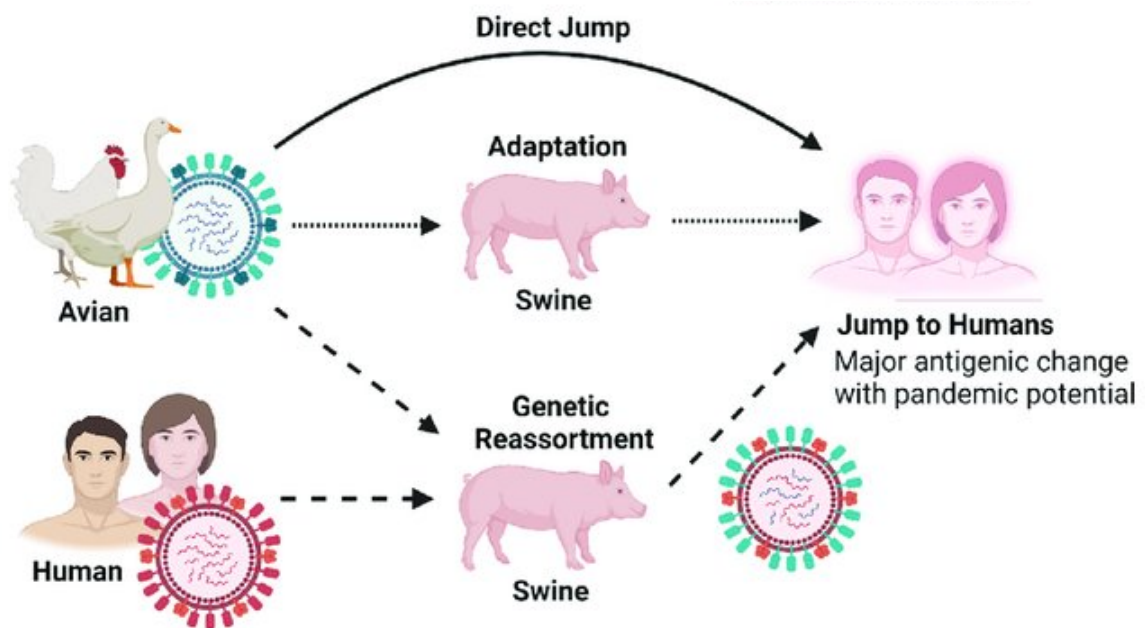


Fig. 1.2

Outline how a new strain may arise due to genetic reassortment.

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..... [3]

- (iv) Suggest how adaptation of influenza virus in swine, as seen in Fig. 1.2, may have occurred.

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..... [2]

- (v) An individual's immune responses can change throughout their lifetime.

Fig. 1.3 shows one person's immune response to the influenza virus when they were first infected and when they were infected two years later by a new, mutated strain of the virus.

The influenza virus has many antigens to which the immune system can respond. Fig. 1.3 shows the response to four of these antigens (A–D).

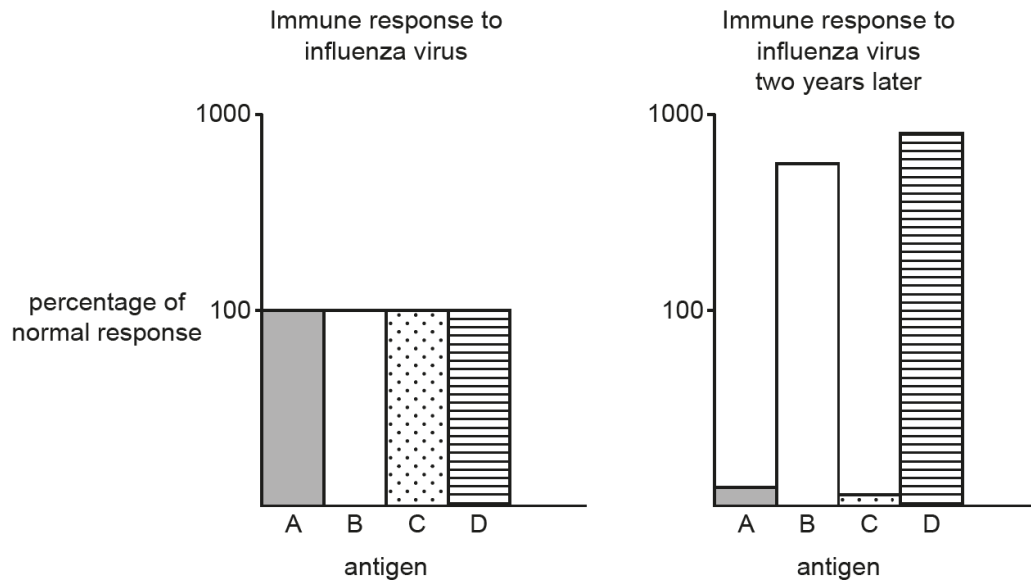


Fig. 1.3

Explain the differences in the person's initial immune response to the influenza virus with their immune response two years later.

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..... [2]

(b) Tuberculosis (TB) is another respiratory infectious disease that affects humans.

The vaccine used to control TB is known as *Bacillus Calmette-Guérin* (BCG). The vaccine contains live bacteria that have been selected so that they do not cause disease in humans.

Fig. 1.4 shows a macrophage that is in the process of engulfing the bacteria in the vaccine.

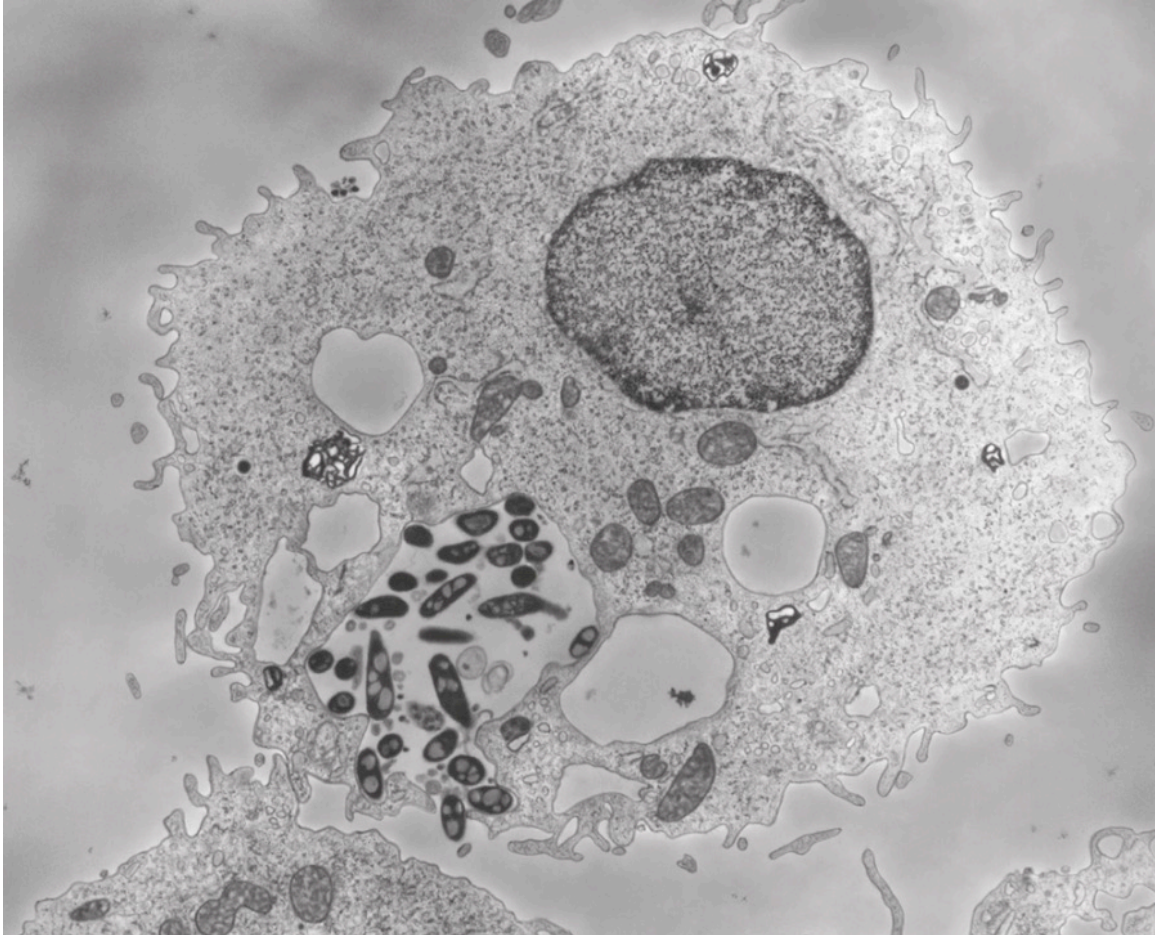


Fig. 1.4

(i) Name the pathogen that causes TB.

..... [1]

(ii) Describe how this pathogen is transmitted.

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..... [2]

- (iii) Outline the events that occur in the body after the macrophage has engulfed the bacteria until the production of antibodies in response to the BCG vaccine.

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..... [5]

- (iv) The treatment for people with active TB lasts six months and involves a combination of antibiotics. This is usually very effective if the person has a susceptible (non-resistant) strain of the bacteria.

Table 1.1 summarises one recommended treatment strategy that involves a combination of antibiotics.

Table 1.1

antibiotic	length of treatment	mode of action of antibiotic
rifampicin (R)	6 months	enters bacterial cells and inhibits protein synthesis
isoniazid (H)	6 months	prevents the synthesis of cell wall components known as mycolic acids
ethambutol (E)	first two months	prevents mycolic acids from being added to the cell wall
pyrazinamide (Z)	first two months	prevents the synthesis of fatty acids

Susceptible strains of the bacteria will be killed using any one of the antibiotics listed in Table 1.1. However, combination treatment is preferred as it is one method that can be used to reduce the impact to society of antibiotic resistance.

With reference to Table 1.1, explain how combination treatment for TB can help to reduce the impact of antibiotic resistance compared to single antibiotic treatment.

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..... [2]

- (v) Rifampicin binds tightly to an RNA polymerase molecule close to its active site. This affects the activity of the enzyme.

During the formation of RNA, a number of events occur that involve the action of RNA polymerase.

Suggest two ways in which rifampicin can affect the activity of RNA polymerase.

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..... [2]

- (vi) Countries are classified by the World Bank into one of four income groups.

Table 1.2 shows the estimated incidence of TB for 2012 to 2016 for these income groups.

The incidence represents the number of new cases of TB occurring per 100 000 people in one year. The new cases include the number of cases that have occurred again after a period of recovery (relapse TB).

Table 1.2

year income group	incidence per 100 000 people				
	2012	2013	2014	2015	2016
low	253	244	238	231	224
lower middle	244	240	236	232	227
upper middle	84	81	78	76	74
high	14	13	13	12	12

Describe the patterns and trends shown in Table 1.2.

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..... [2]

- (vii) In general, the countries that do not have a BCG vaccination programme are high-income countries that have a low number of cases of TB. In most of these countries, the vaccine is given only to babies and children at high risk of developing TB.

Suggest **one** reason why a child in a country with a low number of cases of the disease could be at a high risk of developing TB.

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..... [1]

- (c) Sucrose phosphorylase is an enzyme found in some species of bacteria, including bacteria causing TB. One function of this enzyme is for the production of compounds that help to protect the cell from harmful osmotic changes in the external environment.

- (i) Fig. 1.5 shows the reversible reaction that takes place within the bacterial cell.

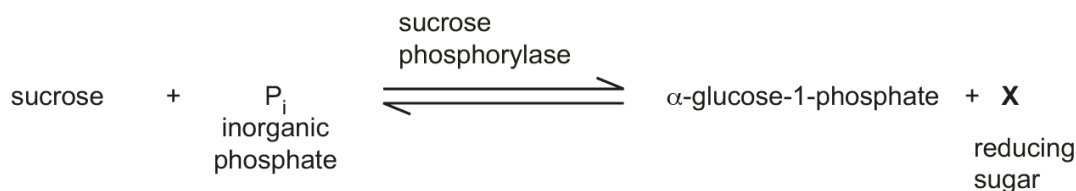


Fig. 1.5

Name reducing sugar **X** in Fig. 1.5.

..... [1]

- (ii) In the absence of sucrose phosphorylase as a catalyst, the reaction shown in Fig. 1.5 would take too long to occur to allow the bacterial cell to function efficiently.

Explain why the reaction shown in Fig. 1.5 proceeds at a much faster rate in the presence of the enzyme.

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..... [1]

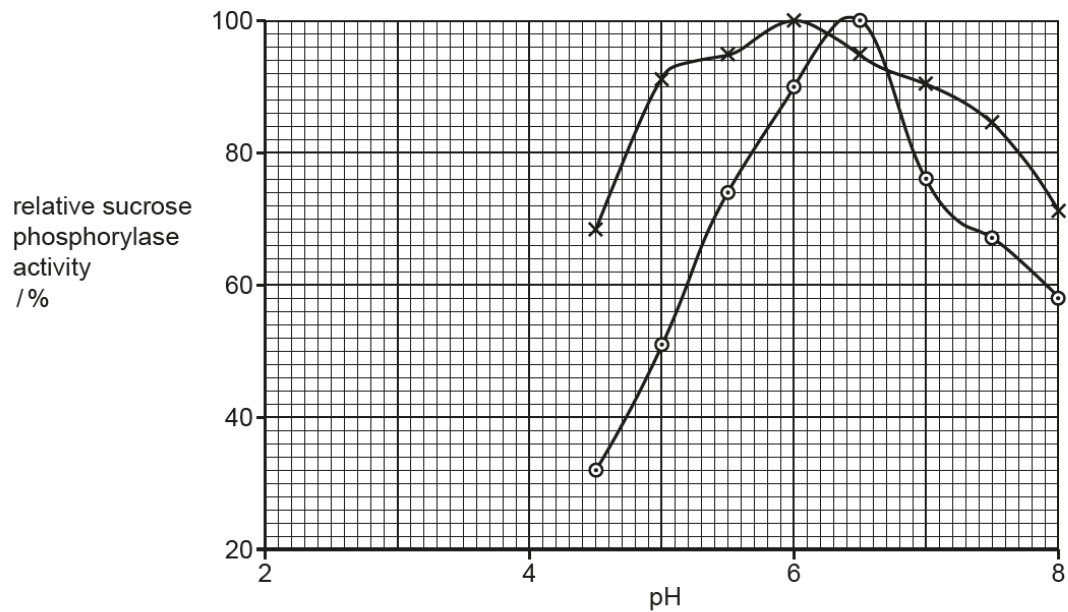
- (iii) An enzyme that catalyses a reaction of commercial interest needs to be investigated to see if it is suitable for use in industry.

For example:

- immobilised enzymes may be used as they have a longer shelf-life than the enzyme free in solution
- many industrial reactions are carried out at higher temperatures to minimize contamination of products by microorganisms.

Fig. 1.6 shows the results of an investigation to compare the activity of sucrose phosphorylase free in solution (free enzyme) with immobilised sucrose phosphorylase (immobilised enzyme) at different pHs.

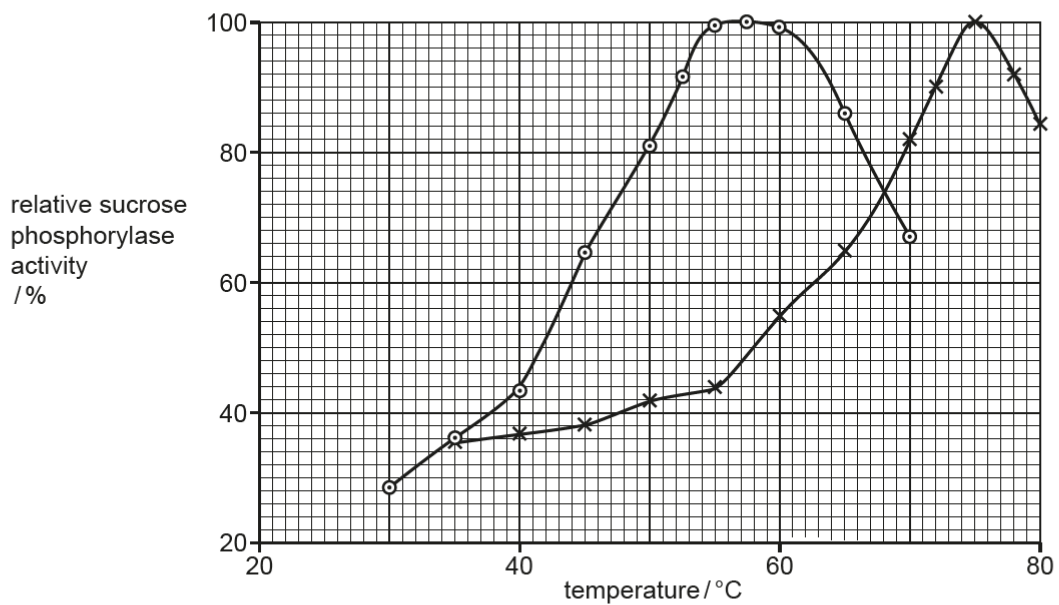
Fig. 1.7 shows the activity of the free enzyme and immobilised enzyme at different temperatures.



Key

- free enzyme
- × immobilised enzyme

Fig. 1.6



Key

- free enzyme
- × immobilised enzyme

Fig. 1.7

With reference to the results shown in Fig. 1.6 and Fig. 1.7, discuss why **immobilised** sucrose phosphorylase enzyme could be better for use in industrial reactions.

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..... [3]

[Total: 30]

- 2 Researchers investigated the extent to which the founder effect and natural selection affected evolutionary change.

Fig. 2.1 shows the brown anole lizard, *Anolis sagrei*. These lizards live on a number of Caribbean islands and feed on a variety of invertebrates and other small animals.

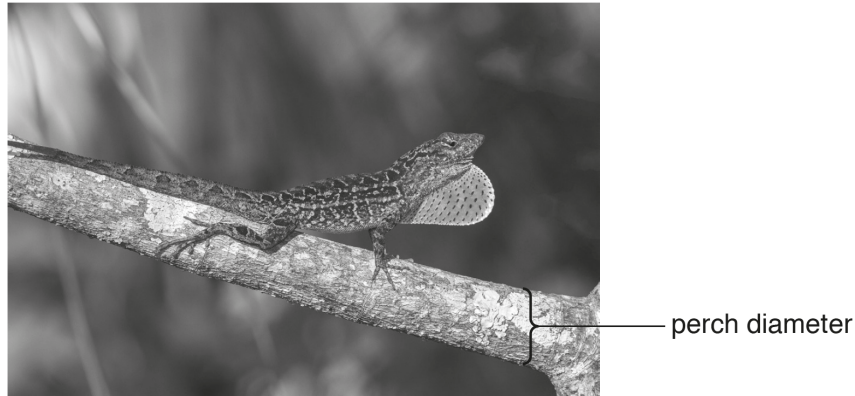


Fig. 2.1

A. sagrei spends a lot of time perching (resting) on, or moving along, branches of shrubs and trees. The width of the branch that *A. sagrei* perches on is known as the perch diameter, as labelled in Fig. 2.1.

There is a positive correlation between perch diameter and hind limb length of *A. sagrei*.

- Longer hind limbs allow *A. sagrei* to run faster on vegetation with a larger diameter.
- Shorter hind limbs are needed to provide stability on vegetation of a smaller diameter.

In 2004, a hurricane caused the death of all the *A. sagrei* lizards on seven islands.

In 2005, the researchers randomly collected seven male and seven female lizards from a source population on a nearby island. For each of the seven islands affected by the hurricane, a male and female lizard were mated and placed on each island. These islands formed the experimental founder islands where new populations of *A. sagrei* were successfully established from each founding pair.

Fig. 2.2 shows the difference in vegetation between the source island and the seven experimental founder islands.

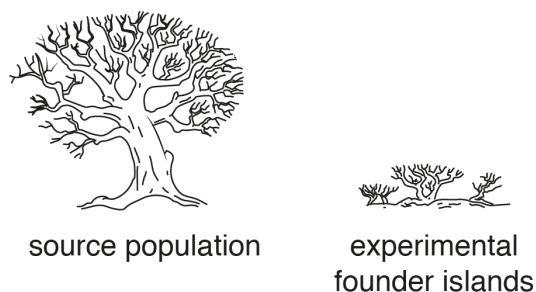


Fig. 2.2

- (a) (i) Predict the effect of natural selection on mean hind limb length of *A. sagrei* on the seven experimental founder islands.

.....
 [1]

- (ii) Predict how collecting individuals at random for the seven founding pairs affects the mean hind limb length of *A. sagrei* on the different islands.

.....
 [1]

- (b) Many generations of *A. sagrei* were produced over the four years after the introduction of the founding pairs.

Fig. 2.3 shows how the mean hind limb length of *A. sagrei* changed on the seven experimental islands and on the source island.

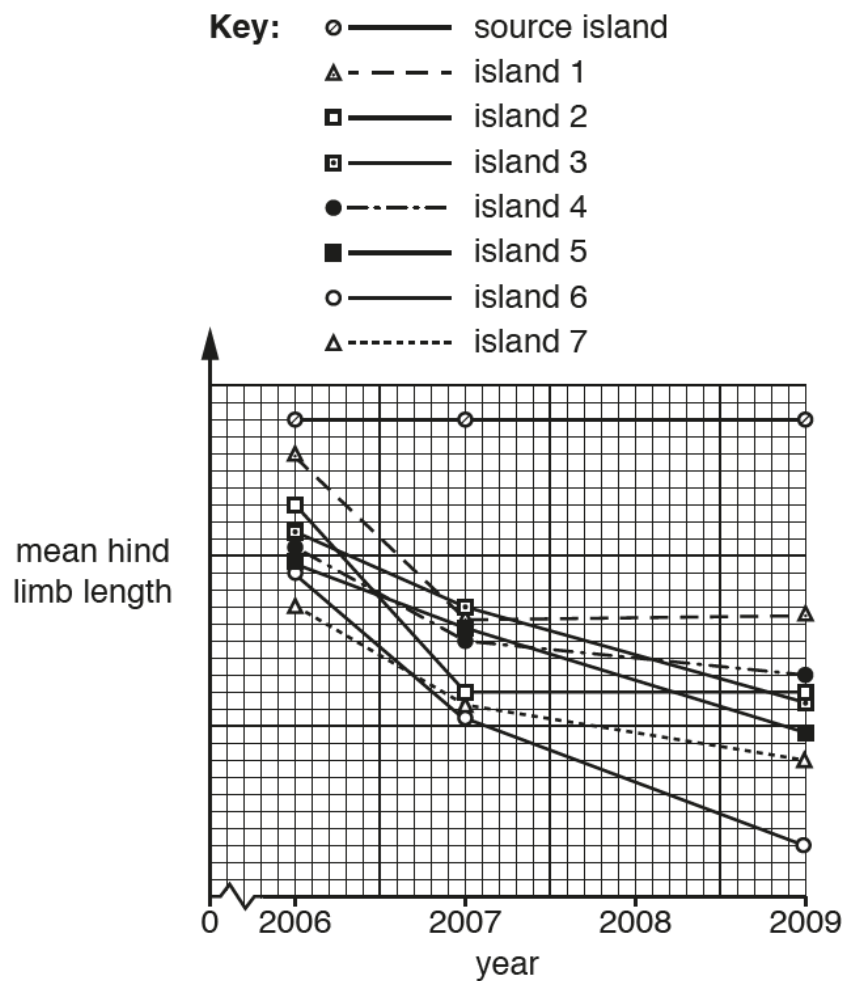


Fig. 2.3

With reference to Fig. 2.2 and Fig. 2.3, describe **and** suggest explanations for the results for the islands.

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..... [5]

- (c) In the investigation, one population of *A. sagrei* was established on each experimental founder island.

Outline how speciation may occur on the seven experimental founder islands.

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..... [3]

[Total: 10]

3 (a) Fig. 3.1 shows the structure of a prokaryotic cell.

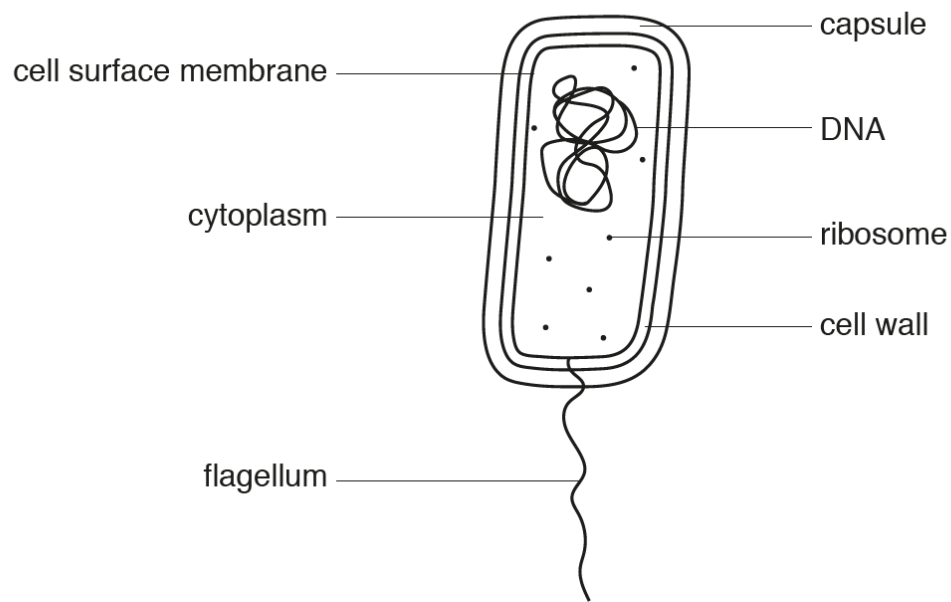


Fig. 3.1

Fig. 3.1 has not been fully labelled to confirm that the cell is prokaryotic.

State what other information could be added to two of the labels to confirm that this cell is prokaryotic and not eukaryotic.

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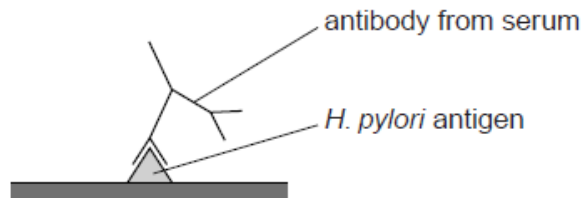
..... [1]

- (b) The bacterium *Helicobacter pylori* has been associated with diseases such as gastric ulcers and stomach cancer.

An infection with *H. pylori* can be diagnosed by testing for the antibodies produced in response to *H. pylori* antigens, as shown in Fig. 3.2.

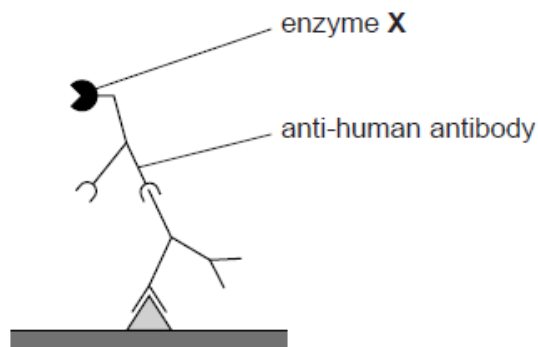
step 1

add blood serum samples to antigens of *H. pylori* attached to wells in a testing plate



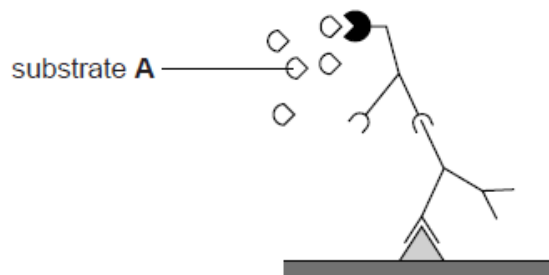
step 2

rinse the testing plate and add anti-human (secondary) antibody linked to enzyme X



step 3

rinse the testing plate and add substrate A, which is converted to a coloured product by enzyme X



step 4

reaction stopped and colour noted

Fig. 3.2

With reference to Fig. 3.2, explain the importance of the following in this test:

(i) the *H. pylori* antigen.

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..... [1]

(ii) the use of enzyme X

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..... [1]

(c) Amoxicillin is an antibiotic that is commonly used to treat *H. pylori*, it is similar to penicillin.

Suggest the mode of action of amoxicillin.

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..... [2]

(d) In some people, *H. pylori* infection can be difficult to treat because, over time, new strains of bacteria arise that have resistance to commonly used antibiotics. The resistance to antibiotics has been found to spread without the need of a vector or bacterium.

Explain how resistance to antibiotics is spread in a population of *H. pylori*.

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..... [3]

- (e) *H. pylori* has an operon that is involved in synthesis of an amino acid, **X**. Fig 3.3 shows the operon and how it is being regulated.

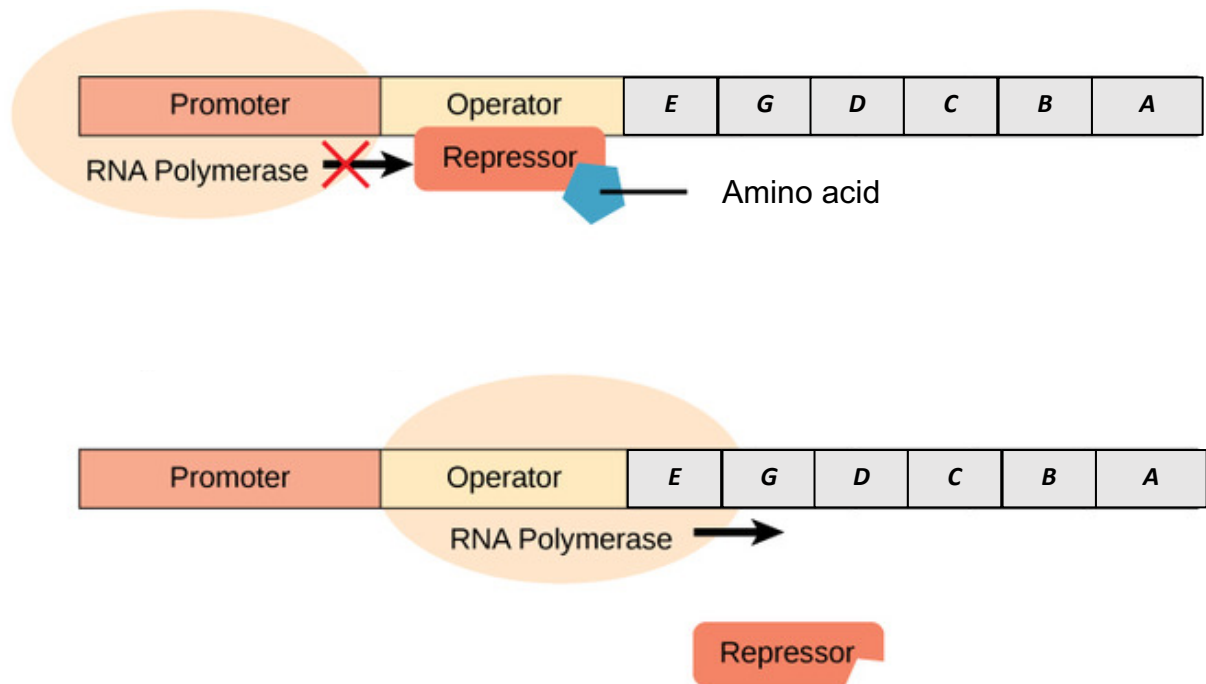


Fig. 3.3

- (i) identify the type of operon.

..... [1]

- (ii) predict the effect of an insertional mutation in the regulatory gene that codes for the repressor.

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..... [1]

[Total: 10]

Section B

Answer **one** question in this section.

Write your answers in the **12-page Answer Booklet** provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)** and **(b)**, as indicated in the question.

- 4 (a)** Molecular biology techniques are common methods used in molecular biology, biochemistry, and genetics which generally involve manipulation and analysis of DNA.

Outline the principles of techniques used to analyse DNA. [10]

- (b)** Cycles play important roles in both natural and man-made biological processes.

Write an essay about cycles in biology. [15]

- 5 (a)** Explain how anatomical and molecular homology support Darwin's theory of evolution on descent with modification. [10]

- (b)** Explain the significance of different biomolecular composition in membranes of different cells and different organelles. [15]

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